

Periodic Steady State

- 1 Systems with Periodic Inputs
 - decay of periodic effects
- 2 The Helmholtz Phasor Method
 - solving the heat equation
 - the phasor solution
- 3 Application
 - the diffusion constant of material

MCS 472 Lecture 38
Industrial Math & Computation
Jan Verschelde, 20 April 2026

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decay of periodic inputs

How deeply within the earth is felt the temperature at the surface?

The temperature at the surface is periodic:

- warm (or even hot) during the day,
- cold when the sun sets and at night.

We have a system with a periodic input.

How long before the periodic inputs have decayed?

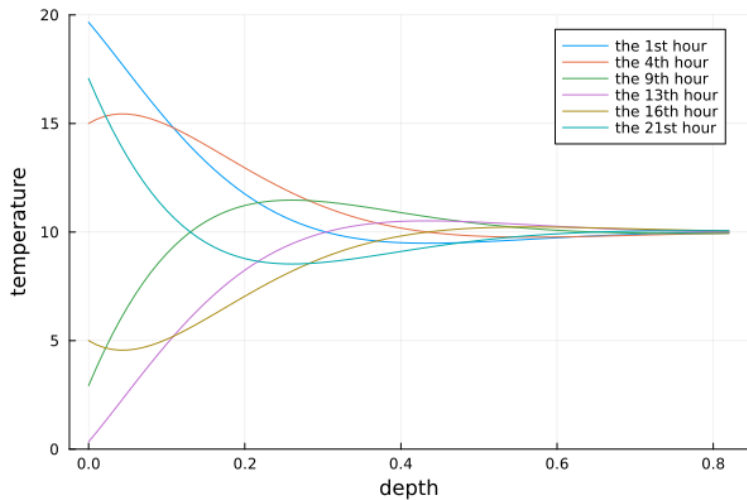
example questions

How deeply within the earth is felt the temperature at the surface?

- Geothermal heat pumps, how deep to bury?
- Heating of concrete, how thick should walls be?
- How deep do worms go to prevent from freezing?

*How deep do we have to dig
to store something at constant temperature?*

a simulation



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the heat equation

The temperature $u(x, t)$

- at depth x ,
- at time t ,

should be a periodic function with decaying amplitude.

Solve the heat equation

$$\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2}$$

with boundary conditions

- $u(0, t) = \sin(\omega t)$ is the temperature at the surface,
- $u(\infty, t) = 0$, the surface temperature is not felt at ∞ .

Our model uses idealized numbers.

a first proposed solution

Our intuition brings up

$$U(x, t) = X(x) \sin(\omega t - \theta(x)),$$

where

- the amplitude $X(x)$ decreases for increasing x ,
- $\theta(x)$ is the phase lag at depth x .

The problem with this proposal is that substituting $U(x, t)$ in the heat equation and then solving for $X(x)$ and $\theta(x)$ is hard.

Exercise 1:

Use `SymPy` to substitute $U(x, t)$ in the heat equation.
Argue why that method does not work.

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the phasor solution

Helmholtz proposed

$$\rho(x, t) = e^{i(\omega t - \theta(x))} X(x) = e^{i\omega t} Y(x),$$

which is called *a phasor solution*.

Substituting $U(x, t) = e^{i\omega t} Y(x)$ into the heat equation will lead to ordinary differential equations.

reduction to ordinary differential equations

In substituting $U(x, t) = e^{i\omega t} Y(x)$ into

$$\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2}$$

we compute

$$\begin{aligned}\frac{\partial U}{\partial t} &= i\omega e^{i\omega t} Y(x) \\ \frac{\partial^2 U}{\partial x^2} &= e^{i\omega t} Y''(x).\end{aligned}$$

This leads to the ordinary differential equation

$$Y''(x) = i\omega Y(x).$$

a symbolic solution

```
julia> using SymPy
julia> x = Sym("x")
julia> omega = Sym("omega")
julia> Y = sympy.Function("y")
julia> ode = diff(Y(x), x, 2) - sympy.I*omega*Y(x)
julia> println(dsolve(ode, Y(x)))
Eq(y(x), C1*exp(-x*sqrt(I*omega))
      + C2*exp(x*sqrt(I*omega)))
```

So sympy computed

$$Y(x) = C_1 e^{-x\sqrt{i\omega}} + C_2 e^{+x\sqrt{i\omega}}, \quad i = \mathbb{I} = \sqrt{-1}.$$

exponential decay

We expect the solution

$$Y(x) = C_1 e^{-x\sqrt{i\omega}} + C_2 e^{+x\sqrt{i\omega}}$$

to decay. Therefore, consider the real and imaginary parts of $\sqrt{i}$

$$\sqrt{i} = \frac{\sqrt{2}}{2} (1 + i),$$

which leads to

$$Y(x) = C_1 e^{-x\frac{\sqrt{2}}{2}(1+i)\sqrt{\omega}} + C_2 e^{+x\frac{\sqrt{2}}{2}(1+i)\sqrt{\omega}}.$$

In our model: $x \geq 0$, and we want decay. Thus: $C_2 = 0$.

To work with idealized temperatures, we normalize $C_1 = 1$.

return to the phasor solution

We found $Y(x) = e^{-x\frac{\sqrt{2}}{2}(1+i)\sqrt{\omega}}$ for use in $U(x, t) = e^{i\omega t} Y(x)$,
where $Y(x)$ appears in the phasor $\rho(x, t) = e^{i(\omega t - \theta(x))} X(x) = e^{i\omega t} Y(x)$.

Rewrite $Y(x)$ into two parts, an amplitude and phase lag part:

$$\begin{aligned} Y(x) &= e^{-x\frac{\sqrt{2}}{2}(1+i)\sqrt{\omega}} \\ &= e^{-x\frac{\sqrt{2}}{2}\sqrt{\omega}} e^{-xi\frac{\sqrt{2}}{2}\sqrt{\omega}} \\ &= \left(\underbrace{e^{-x\sqrt{\omega/2}}}_{\text{decays}} \right) \left(\underbrace{e^{-xi\sqrt{\omega/2}}}_{\text{oscillates}} \right) \\ &= \underbrace{X(x)}_{\text{amplitude}} \cdot \underbrace{e^{-i\theta(x)}}_{\text{phase lag}} \end{aligned}$$

the phasor solution

The phasor solution is

$$\rho(x, t) = e^{i\omega t} Y(x),$$

with

$$Y(x) = \left(e^{-x\sqrt{\omega/2}} \right) \left(e^{-xi\sqrt{\omega/2}} \right).$$

Thus, the phasor solution is

$$\rho(x, t) = \left(e^{-x\sqrt{\omega/2}} \right) \left(e^{i(\omega t - x\sqrt{\omega/2})} \right).$$

the heat equation with an input function

Exercise 2:

Apply the Helmholtz phasor method to solve

$$\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2}, \quad x \in [0, 1],$$

subject to the boundary conditions:

- $u(0, t) = \sin(t)$,
- $u(1, t) = 0$.

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the diffusion constant of material

Let α be the diffusion constant of material.

Introduce α in the phasor solution

$$\rho(x, t) = \left(e^{-x\sqrt{\omega/(2\alpha)}} \right) \left(e^{i(\omega t - x\sqrt{\omega/(2\alpha)})} \right).$$

The *depth of penetration* is $\delta = \sqrt{\frac{2\alpha}{\omega}}$, the reciprocal of $\sqrt{\omega/(2\alpha)}$.

If $x = \delta$, then the amplitude is $e^{-1} \approx 0.3678$.

The depth of penetration is the depth at which amplitude variations have dropped to $e^{-1} \approx 37\%$ of the surface temperature.

our mathematical model

Using $\delta = \sqrt{\frac{2\alpha}{\omega}}$ in $\rho(x, t) = \left(e^{-x\sqrt{\omega/(2\alpha)}}\right) \left(e^{i(\omega t - x\sqrt{\omega/(2\alpha)})}\right)$ gives,

if we focus on the real part:

$$T(x, t) = T_0 + Ae^{-x/\delta} \cos\left(\omega t - \frac{x}{\delta}\right),$$

where

- T_0 is the temperature at the surface, and
- A is the maximal variation of the surface temperature.

a numerical simulation

The parameters for a numerical simulation are

$$\alpha = 1.0\text{E}-6$$

$$P = 24 \cdot 3600$$

$$\omega = 2\pi/P$$

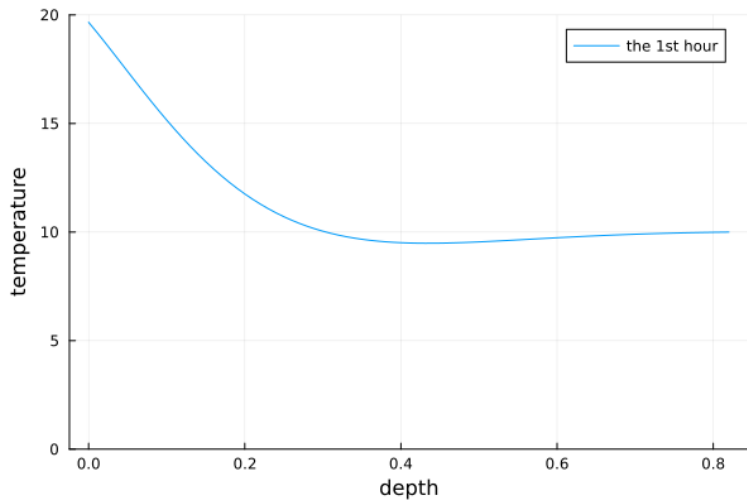
$$T_0 = 10$$

$$A = 10$$

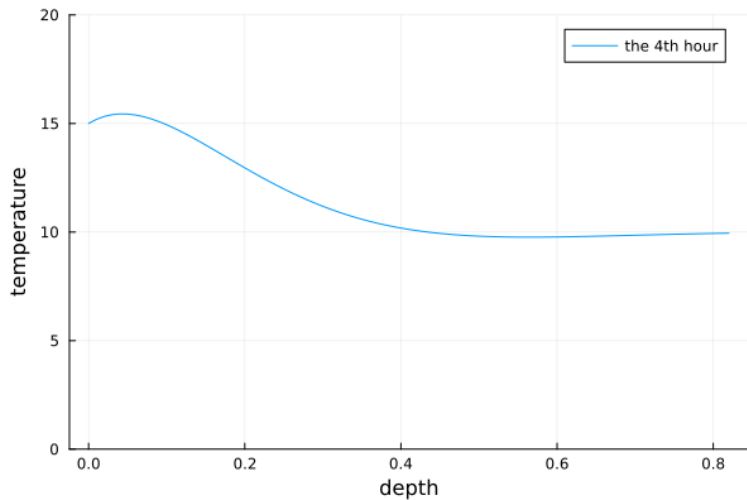
For P , we take a 24-hour period, measured in seconds.

See the posted Jupyter notebook.

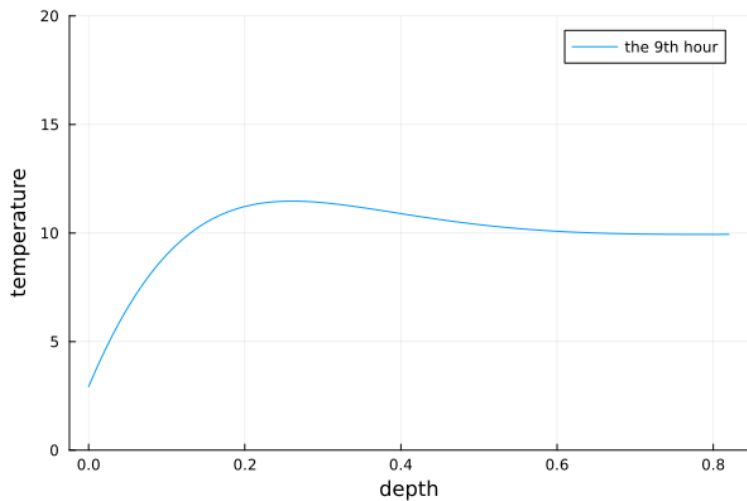
after one hour



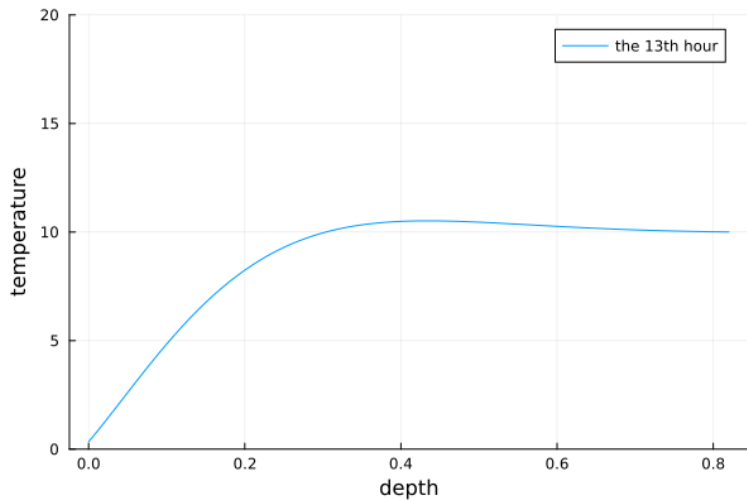
after four hours



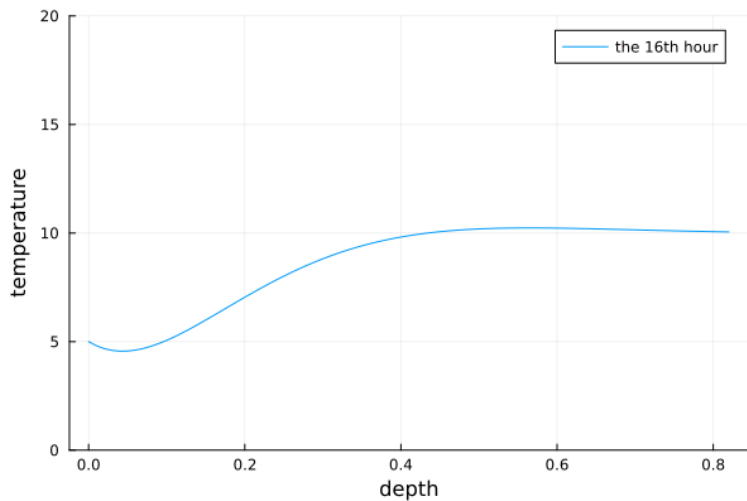
after nine hours



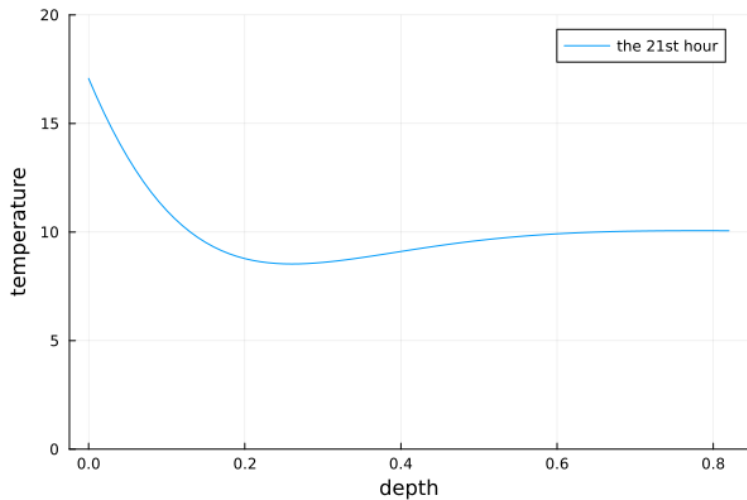
after thirteen hours



after sixteen hours



after twenty one hours



reduction to one parameter

Exercise 3: We derived

$$T(x, t) = T_0 + Ae^{-x/\delta} \cos\left(\omega t - \frac{x}{\delta}\right).$$

Show that by the substitutions $(T(x, t) - T_0)/A = \widehat{T}$ and $\omega t = \widehat{t}$, the model becomes

$$\widehat{T}(x, \widehat{t}) = e^{-x/\delta} \cos\left(\widehat{t} - \frac{x}{\delta}\right),$$

and thus dependent only on the penetration depth δ ,
to answer our original question:

*How deep should we bury an object
to shield it from temperature fluctuations?*

geothermal heat pumps

A geothermal heat pump compensates for the seasonal temperature fluctuations, to heat a home during the winter with the warmth of the earth stored during summer.

Proposal for a Topic of a Project:

- Use the Helmholtz phasor method to explain how deep the system must be buried.
- Do a cost benefit analysis of installing a geothermal heat pump, comparing to heating a home with a furnace.

thickness of walls for isolation

Native Americans of the southwest built homes with adobe walls of a critical thickness. The walls delayed the flux of the sun until it was needed to heat the interior during the cold nights.

Proposal for a Topic of a Project:

- Model the thickness of the adobe walls.
- Use actual material properties to select the optimal thickness for a 24-hour repetitive cycle.
- Compare the results of your model against actual constructions.

study hibernating worms

Birds migrate to survive the winter.

Worms have no wings but they do survive each winter.

Proposal for a Topic of a Project:

How deep do hibernating worms go to survive the winter?

- Gather biological data on worms.
- How deep does one have to dig to find worms?
- Use the Helmholtz method to answer the question.

summary and bibliography

We applied the Helmholtz phasor method to study systems driven by sinusoidal inputs.

Periodic steady state is covered in section 11.5 of our text book.

- Charles R. MacCluer:
Industrial Mathematics. Modeling in Industry, Science, and Government. Prentice Hall, 2000.